

Category Theory and Functional Programming

§ 1.6 The Curry-Howard Correspondence

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Preliminaries

Convention

The number assigned to chapters, examples, exercises, figures, pages, sections, and theorems on these slides correspond to the numbers assigned in the textbook [Abramsky and Tzevelekos 2011].

Outline

Typed Lambda-Calculus

References

Typed Lambda-Calculus

Definition

A **typed λ -calculus** with function types, product types, unit type and empty type, is defined by:

(i) Types

$\sigma, \tau ::= b$	(base types)
$\sigma \rightarrow \tau$	(function types)
$\sigma \times \tau$	(product types)
1	(unit type)
0	(empty type)

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(ii) Terms

$M, N ::= x$	(variable)
$\lambda x.M$	(λ -abstraction)
$M N$	(application)
$\langle M, N \rangle$	(pair)
$\pi_1 M \mid \pi_2 M$	(projections)
$\langle \rangle$	(unit)
absurd M	(absurd)

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(iv) Conversion rules

$$\lambda x.M = \lambda y.M[x := y] \quad (\alpha\text{-conversion})$$

$$(\lambda x.M) N = M[x := N] \quad (\beta\text{-conversion})$$

$$\lambda x.M x = M, \text{ with } x \text{ not in } M \quad (\eta\text{-conversion})$$

$$\pi_i \langle M_1, M_2 \rangle = M_i$$

$$M = \langle \pi_1 M, \pi_2 M \rangle$$

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(iii) Contexts

$\Gamma ::= \emptyset$

(empty context)

| $\Gamma, x : \sigma$, with x not in Γ

(context extension)

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Typed Lambda-Calculus

(v) Typing rules

► Hypothesis

$$\frac{}{\Gamma, x : \sigma \vdash x : \sigma} \text{hyp} \quad (\text{if } x : \sigma \in \Gamma)$$

► Function types

$$\frac{\Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x. M : \sigma \rightarrow \tau} \rightarrow\text{I}$$

$$\frac{\Gamma \vdash M : \sigma \rightarrow \tau \quad \Gamma \vdash N : \sigma}{\Gamma \vdash M N : \tau} \rightarrow\text{E}$$

► Product types

$$\frac{\Gamma \vdash M : \sigma \quad \Gamma \vdash N : \tau}{\Gamma \vdash \langle M, N \rangle : \sigma \times \tau} \times\text{I}$$

$$\frac{\Gamma \vdash M : \sigma_1 \times \sigma_2}{\Gamma \vdash \pi_i M : \sigma_i} \times\text{E}$$

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Typed Lambda-Calculus

(v) Typing rules

► Unit type

$$\frac{}{\Gamma \vdash \langle \rangle : 1} \text{1I}$$

► Empty type

$$\frac{\Gamma \vdash M : 0}{\Gamma \vdash \text{absurd } M : \sigma} \text{0E}$$

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Typed Lambda-Calculus

Category associated to the typed λ -calculus

The category **TLC** is defined by:

- (i) Objects: Types
- (ii) Arrows: There is an arrow $\sigma \xrightarrow{M} \tau$ iff $M : \sigma \rightarrow \tau$.

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(iii) Composition

If $M : \rho \rightarrow \sigma$ and $N : \sigma \rightarrow \tau$ then

$$M \circ N : \rho \rightarrow \tau$$

$$M \circ N := \lambda x. M (N x)$$

(iv) Identity arrows

$$\text{id}_\sigma : \sigma \rightarrow \sigma$$

$$\text{id}_\sigma := \lambda x. x, \text{ with } x : \sigma$$

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(v) Unit axiom

Let $M : \sigma \rightarrow \tau$, then

$$\begin{aligned} M \circ \text{id}_\sigma &= \lambda x. M(\text{id}_\sigma x) \\ &= \lambda x. M((\lambda y. y) x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

$$\begin{aligned} \text{id}_\tau \circ M &= \lambda x. \text{id}_\tau (M x) \\ &= \lambda x. (\lambda y. y) (M x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

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(vi) Associativity axiom

Let $M : \sigma_1 \rightarrow \sigma_2$, $N : \sigma_2 \rightarrow \sigma_3$ and $P : \sigma_3 \rightarrow \sigma_4$. Then

$$\begin{aligned} M \circ (N \circ P) &: \sigma_1 \rightarrow \sigma_4 \\ M \circ (N \circ P) &= \lambda x. M ((N \circ P) x) \\ &= \lambda x. M ((\lambda y. N (P y)) x) \\ &= \lambda x. M (N (P x)) \\ &= \lambda x. (\lambda y. M (N y)) (P x) \\ &= \lambda x. (M \circ N) (P x) \\ &= (M \circ N) \circ P \end{aligned}$$

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(vii) Initial object: Type 0

$$i_\sigma : 0 \rightarrow \sigma$$

$$i_\sigma := \text{absurd}$$

(viii) Terminal object: Type 1

$$t_\sigma : \sigma \rightarrow 1$$

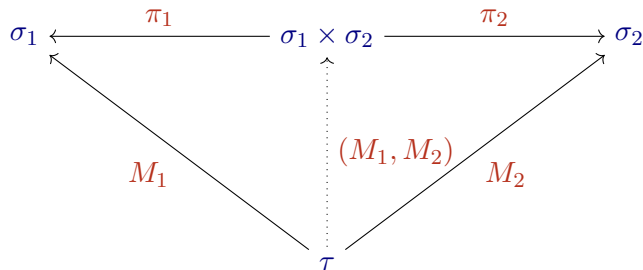
$$t_\sigma := \lambda x. \langle \rangle$$

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Typed Lambda-Calculus

(ix) Binary products

We start with the objects and the arrows in the following diagram



where $(M_1, M_2) := \lambda x. \langle M_1 x, M_2 x \rangle : \tau \rightarrow \sigma_1 \times \sigma_2$,

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First, we prove that the diagram is commutative.

$$\begin{aligned}\pi_i \circ (M_1, M_2) &: \tau \rightarrow \sigma_i \\ \pi_i \circ (M_1, M_2) &= \pi_i \circ \lambda y. \langle M_1 y, M_2 y \rangle \\ &= \lambda x. \pi_i ((\lambda y. \langle M_1 y, M_2 y \rangle) x) \\ &= \lambda x. \pi_i \langle M_1 x, M_2 x \rangle \\ &= \lambda x. M_i x \\ &= M_i\end{aligned}$$

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Next, we prove the uniqueness of arrow (M_1, M_2) . Let $N : \sigma_1 \times \sigma_2$ such that $\pi_i \circ N = M_i$. Then

$$\begin{aligned}(M_1, M_2) &= \lambda x. \langle M_1 x, M_2 x \rangle \\ &= \lambda x. \langle (\pi_1 \circ N) x, (\pi_2 \circ N) x \rangle \\ &= \lambda x. \langle (\lambda y. \pi_1 (N y)) x, (\lambda y. \pi_2 (N y)) x \rangle \\ &= \lambda x. \langle \pi_1 (N x), \pi_2 (N x) \rangle \\ &= \lambda x. N x \\ &= N\end{aligned}$$

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(x) Finitary products

Since **TLC** has binary products and a terminal object (nullary product) then it has finite products.

References

References



Abramsky, S. and Tzevelekos, N. (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Coecke, B. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).